

Multi-Tier Enterprise Campus Network Portfolio

Project Overview

I designed and built a highly available, secure, three-tier enterprise campus network to simulate a production corporate environment. The network architecture transitions from a redundant Multilayer Switch distribution core down to a segmented Access Layer supporting automated client onboarding and voice prioritization. The design integrates dynamic routing, default gateway redundancy, automated IP provisioning, voice services, and strict security hardening to protect the corporate infrastructure from the core down to the user switchports.

Design Breakdown & Implementation Decisions

1. Core Routing & High Availability

OSPFv2 Dynamic Routing

I configured OSPFv2 as the interior gateway protocol across the backbone network, assigning unique router IDs (1.1.1.1 on **MSW1** and 2.2.2.2 on **MSW2**). By binding the local subnets directly into Area 0, the distribution switches dynamically exchange paths with the upstream core layer. This removes manual static routing overhead and ensures the network can automatically calculate and converge on optimal data paths if a transit link fails.

```
MSW1#show ip route ospf
 10.0.0.0/30 is subnetted, 4 subnets
O    10.0.0.0 [110/2] via 10.0.0.6, 00:07:52, GigabitEthernet0/1
O    10.0.0.8 [110/2] via 192.168.10.3, 00:00:07, Vlan10
      [110/2] via 192.168.20.3, 00:00:07, Vlan20
      [110/2] via 192.168.30.3, 00:00:07, Vlan30
      [110/2] via 192.168.40.3, 00:00:07, Vlan40
 172.16.0.0/30 is subnetted, 1 subnets
O    172.16.0.0 [110/66] via 10.0.0.6, 00:07:52, GigabitEthernet0/1
O    192.168.50.0 [110/67] via 10.0.0.6, 00:07:52, GigabitEthernet0/1
 209.165.200.0/30 is subnetted, 2 subnets
O    209.165.200.0 [110/2] via 10.0.0.6, 00:07:52, GigabitEthernet0/1
O    209.165.200.4 [110/3] via 10.0.0.6, 00:00:07, GigabitEthernet0/1
      [110/3] via 192.168.10.3, 00:00:07, Vlan10
      [110/3] via 192.168.20.3, 00:00:07, Vlan20
      [110/3] via 192.168.30.3, 00:00:07, Vlan30
      [110/3] via 192.168.40.3, 00:00:07, Vlan40
MSW1#
```

MSW1 OSPF routing table displaying learned inter-network paths via GigabitEthernet0/1 and local Switched Virtual Interfaces (SVIs).

```

MSW2#show ip route ospf
 10.0.0.0/30 is subnetted, 3 subnets
O    10.0.0.0 [110/2] via 10.0.0.10, 00:10:03, GigabitEthernet0/1
O    10.0.0.4 [110/2] via 192.168.10.2, 00:02:17, Vlan10
      [110/2] via 192.168.20.2, 00:02:17, Vlan20
      [110/2] via 192.168.30.2, 00:02:17, Vlan30
      [110/2] via 192.168.40.2, 00:02:17, Vlan40
 172.16.0.0/30 is subnetted, 1 subnets
O    172.16.0.0 [110/67] via 10.0.0.10, 00:02:17, GigabitEthernet0/1
      [110/67] via 192.168.10.2, 00:02:17, Vlan10
      [110/67] via 192.168.20.2, 00:02:17, Vlan20
      [110/67] via 192.168.30.2, 00:02:17, Vlan30
      [110/67] via 192.168.40.2, 00:02:17, Vlan40
O    192.168.50.0 [110/68] via 10.0.0.10, 00:02:17, GigabitEthernet0/1
      [110/68] via 192.168.10.2, 00:02:17, Vlan10
      [110/68] via 192.168.20.2, 00:02:17, Vlan20
      [110/68] via 192.168.30.2, 00:02:17, Vlan30
      [110/68] via 192.168.40.2, 00:02:17, Vlan40
 209.165.200.0/30 is subnetted, 2 subnets
O    209.165.200.0 [110/3] via 10.0.0.10, 00:02:17, GigabitEthernet0/1
      [110/3] via 192.168.10.2, 00:02:17, Vlan10
      [110/3] via 192.168.20.2, 00:02:17, Vlan20
      [110/3] via 192.168.30.2, 00:02:17, Vlan30
      [110/3] via 192.168.40.2, 00:02:17, Vlan40
O    209.165.200.4 [110/2] via 10.0.0.10, 00:10:03, GigabitEthernet0/1

```

MSW2#4

MSW2 OSPF routing table verifying path symmetry across the distribution core backbone.

HSRP Gateway Redundancy

To eliminate single points of failure for user endpoints, I deployed HSRP across all user VLANs. I configured **MSW1** with an elevated priority of **110** and preemption enabled to force it as the primary active gateway, while **MSW2** retains the default priority of **100** as the standby peer. End-user devices point to a shared virtual gateway IP (.1). If **MSW1** experiences an outage, **MSW2** transparently assumes control of the virtual IP to maintain continuous uptime for user data and VoIP streams.

```

MSW1#show standby brief
                P indicates configured to preempt.
                |
Interface      Grp  Pri P State      Active      Standby      Virtual IP
Vl10           10   110 P Active     local       192.168.10.3  192.168.10.1
Vl20           20   110 P Active     local       192.168.20.3  192.168.20.1
Vl30           30   110 P Active     local       192.168.30.3  192.168.30.1
Vl40           40   110 P Active     local       192.168.40.3  192.168.40.1
MSW1#$

```

HSRP status on MSW1 confirming its role as the 'Active' default gateway for VLANs 10, 20, 30, and 40.

```

MSW2#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State   Active        Standby        Virtual IP
Vl10        10   100 P Standby  192.168.10.2   local          192.168.10.1
Vl20        20   100 P Standby  192.168.20.2   local          192.168.20.1
Vl30        30   100 P Standby  192.168.30.2   local          192.168.30.1
Vl40        40   100 P Standby  192.168.40.2   local          192.168.40.1
MSW2#

```

HSRP status on MSW2 showing its 'Standby' designation, ready to assume the virtual IP bindings if MSW1 fails.

LACP EtherChannel Link Aggregation

To provide high-bandwidth, redundant cross-links within the core, I bundled physical interfaces into a logical Port-Channel using the Link Aggregation Control Protocol (LACP). Operating in active mode on both ends, interfaces **Fa0/2** and **Fa0/3** negotiate a single logical link (**Po1**). This prevents Spanning Tree Protocol (STP) from placing the redundant cable into a discarding state, effectively doubling throughput while ensuring sub-second link failover.

```

MSW1#show etherchannel summary
Flags: D - down        P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3      S - Layer2
       U - in use      f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP        Fa0/2(P) Fa0/3(P)
MSW1#

```

MSW1 EtherChannel summary displaying Port-channel1 operational as a Layer 2 bundle in use (SU) with ports Fa0/2 and Fa0/3 bundled in-channel (P).

```

MSW2#show etherchannel summary
Flags: D - down        P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3      S - Layer2
       U - in use      f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP        Fa0/2(P) Fa0/3(P)
MSW2#

```

MSW2 EtherChannel verification confirming symmetrical LACP bundling with its distribution neighbor.

2. Infrastructure Services & Automation

Centralized Multi-Scope DHCP & Voice Integration

I turned **MSW1** into a central DHCP server by building independent address pools tailored to the *Management, Sales, Users, and Voice* segments. Explicit exclusion blocks (.1 through .5) protect critical static infrastructure assignments from IP address duplication conflicts.

For the Voice network (VLAN 40), I incorporated **DHCP Option 150**. When Cisco IP Phones boot up, they lease an address within the **192.168.40.0/24** subnet and automatically receive the TFTP server IP (**10.0.0.14**) via Option 150 to fetch their configuration profiles and register for dial tones.

```
MSW1#show ip dhcp binding
```

IP address	Client-ID/ Hardware address	Lease expiration	Type
192.168.10.7	00D0.BAAD.A903	--	Automatic
192.168.40.15	000C.8598.5203	--	Automatic
192.168.40.13	0007.EC6B.1AB1	--	Automatic
192.168.40.16	0003.E4A9.8628	--	Automatic
192.168.40.17	0005.5EA6.7E06	--	Automatic
192.168.40.18	0001.9710.D558	--	Automatic
192.168.40.19	0006.2A3A.6DCD	--	Automatic
192.168.40.22	0040.0B23.5D1E	--	Automatic
192.168.40.24	0030.A390.CBBE	--	Automatic
192.168.40.25	0002.4AE9.8845	--	Automatic

```
MSW1#
```

MSW1 active DHCP bindings showing automated address allocation for IP phones and workstations across subnets.

PAT Boundary Translation

I configured Port Address Translation (PAT) on the perimeter ISP router to map internal private IP space to a single public-facing interface. This architecture secures internal infrastructure addressing profiles from exposure to public segments while enabling external internet connectivity for all local campus VLANs.

3. Device Hardening & Security

Management Plane Protection (SSHv2 & ACLs)

To secure the switch management architecture against credential sniffing, I disabled clear-text Telnet and implemented encrypted **SSHv2**. This configuration utilizes a 2048-bit RSA key pair generated under a dedicated local domain framework ([rachel.fowler](#)).

To enforce the Principle of Least Privilege, I attached an inbound Standard Access Control List (**ACL 10**) directly to the VTY lines (**0 15**). This locks down remote terminal access avenues exclusively to the authorized administrative subnet host ([192.168.50.2](#)), dropping all unauthorized connection attempts at the management boundary.

```
MSW1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
MSW1#show access-lists
Standard IP access list 10
  10 permit host 192.168.50.2
  20 deny any (16 match(es))
MSW1#
```

Cryptographic engine verification confirming SSHv2 deployment alongside the active ruleset for Standard IP Access List 10.

Attack Surface Reduction & Switchport Security

To mitigate physical layer intrusion vectors, such as unauthorized users connecting rogue hardware to unoccupied office wall jacks, I instituted an explicit interface black-hole policy. Every physical switchport not actively connected to an authorized network asset is administratively **shut down** and re-assigned out of the default space into an isolated, non-routed Native VLAN (**VLAN 99**).

Furthermore, access layer boundaries utilize **STP Port-Fast** to bypass legacy transient states for immediate endpoint connectivity, alongside **Port-Security** parameters to control MAC address thresholds per interface.

```

MSW1#show int status
Port      Name      Status      Vlan      Duplex  Speed Type
-----
Po1       connected trunk       auto      auto    10/100BaseTX
Fa0/1     connected trunk       auto      auto    10/100BaseTX
Fa0/2     connected trunk       auto      auto    10/100BaseTX
Fa0/3     connected trunk       auto      auto    10/100BaseTX
Fa0/4     connected trunk       auto      auto    10/100BaseTX
Fa0/5     connected routed      auto      auto    10/100BaseTX
Fa0/6     disabled 99      auto      auto    10/100BaseTX
Fa0/7     disabled 99      auto      auto    10/100BaseTX
Fa0/8     disabled 99      auto      auto    10/100BaseTX
Fa0/9     disabled 99      auto      auto    10/100BaseTX
Fa0/10    disabled 99      auto      auto    10/100BaseTX
Fa0/11    disabled 99      auto      auto    10/100BaseTX
Fa0/12    disabled 99      auto      auto    10/100BaseTX
Fa0/13    disabled 99      auto      auto    10/100BaseTX
Fa0/14    disabled 99      auto      auto    10/100BaseTX
Fa0/15    disabled 99      auto      auto    10/100BaseTX
Fa0/16    disabled 99      auto      auto    10/100BaseTX
Fa0/17    disabled 99      auto      auto    10/100BaseTX
Fa0/18    disabled 99      auto      auto    10/100BaseTX
Fa0/19    disabled 99      auto      auto    10/100BaseTX
Fa0/20    disabled 99      auto      auto    10/100BaseTX
Fa0/21    disabled 99      auto      auto    10/100BaseTX
Fa0/22    disabled 99      auto      auto    10/100BaseTX
Fa0/23    disabled 99      auto      auto    10/100BaseTX
Fa0/24    disabled 99      auto      auto    10/100BaseTX
Gig0/1    connected routed      auto      auto    10/100BaseTX
Gig0/2    disabled 99      auto      auto    10/100BaseTX
MSW1#

```

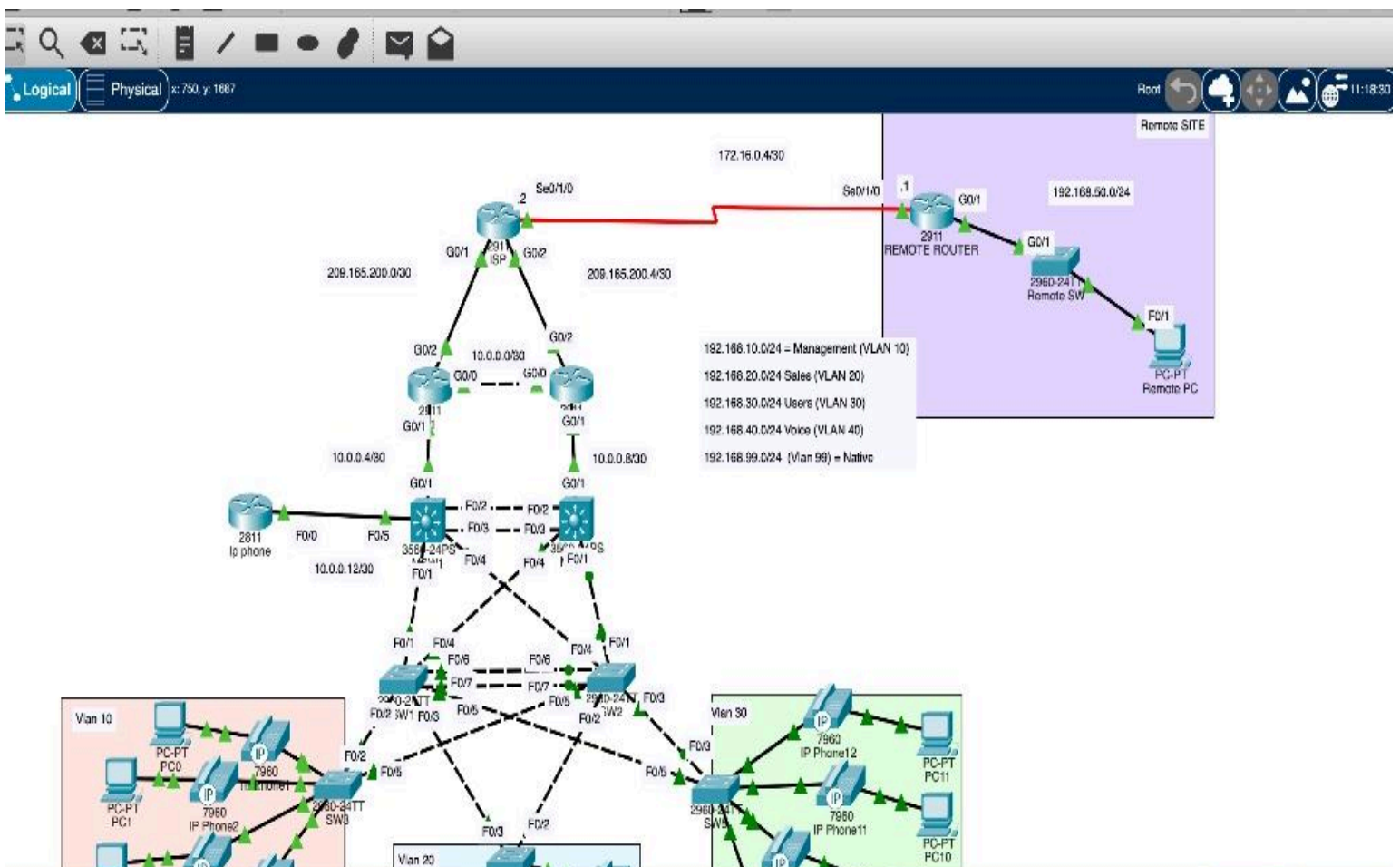
MSW1 interface status map displaying operational trunk/routed interfaces alongside disabled access interfaces securely parked in VLAN 99.

Administrative Baseline

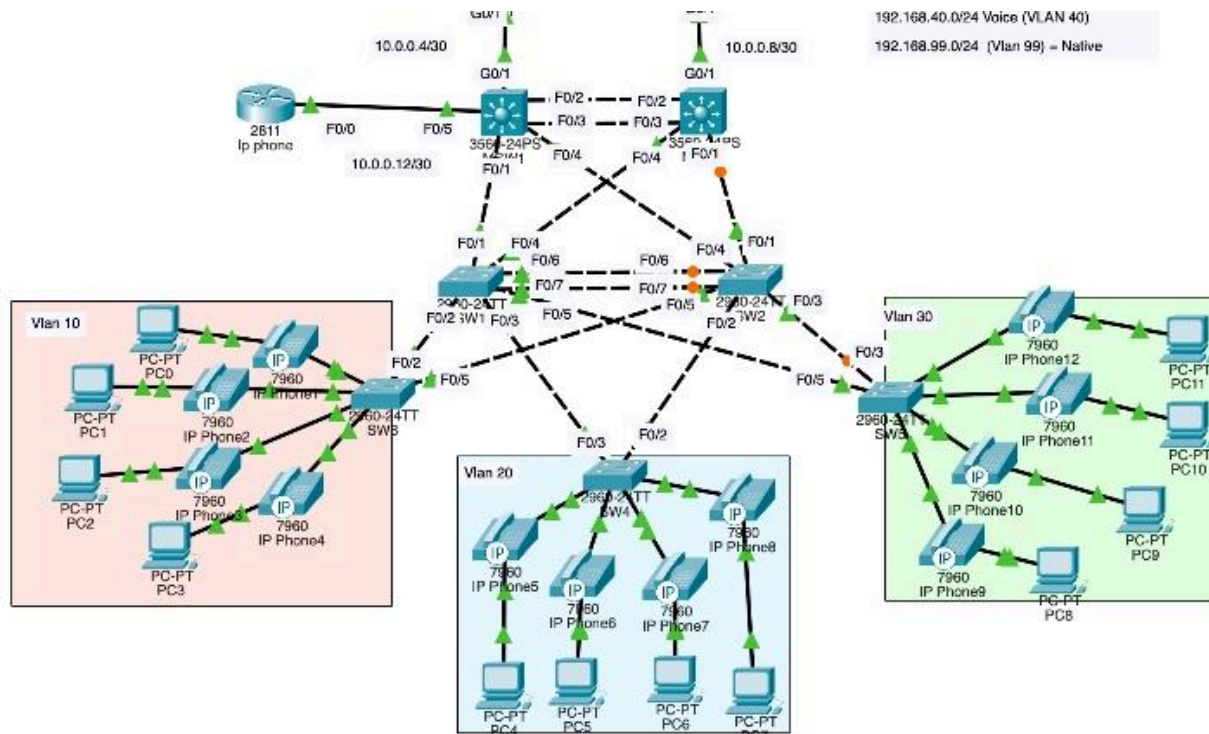
I established custom device hostnames, legal MOTD login banners, and strong **enable secret** password hashes to protect privileged EXEC mode access across all infrastructure components.

4. Network Topology Reference Maps

** Due to the scale of the campus layout, the topology diagram has been split into two detailed sectors to maintain high-resolution visibility for all interface labels and device boundaries.



Displays the redundant multilayer switch core (MSW1/MSW2), transit networks, the simulated ISP infrastructure, and the Serial WAN connection link leading to the remote branch router.



Displays the end-user access layer, including the dual-VLAN segmentation for IP Phones and data PCs across VLANs 10, 20, and 30.